“Well,” replied Alice, “just think about ordinary textbook quantum mechanics. Given a quantum state, we can calculate the total energy it describes. As long as the wave function evolves strictly according to the Schrödinger equation, that energy is exactly conserved, right?”

“Sure.”

“That’s it. In Many-Worlds, the wave function obeys the Schrödinger equation, which conserves energy.”

“But what about the extra worlds?” her father insisted. “I could measure the energy contained in this world I see around me, and you say it’s being duplicated all the time.”

Alice felt she was on firm ground with this one. “Not all worlds are created equal. Think about the wave function. When it describes multiple branched worlds, we can calculate the total amount of energy by adding up the amount of energy in each world, times the weight (the amplitude squared) for that world. When one world divides in two, the energy in each world is basically the same as it previously was in the single world (as far as anyone living inside is concerned), but their contributions to the total energy of the wave function of the universe have divided in half, since their amplitudes have decreased. Each world got a bit thinner, although its inhabitants can’t tell any difference.”

“Mathematically I see what you’re saying,” admitted her father. “But I seem to be lacking some intuition here. I have, say, a bowling ball, with a certain mass and potential energy. But then someone in the next room observes a quantum spin and branches the wave function. Now there are two bowling balls, each of which has the energy of the previous one. No?”

“That ignores the amplitudes of the branches. The contribution of the bowling ball to the energy of the universe isn’t just the mass and the potential energy of the ball; it’s that, times the weight of its branch of the wave function. After the splitting it looks like you have two bowling balls, but together they contribute exactly as much to the energy of the wave function as the single bowling ball did before.”

Her father seemed to ponder this. “I’m not sure I agree with you, but I think you’re wearing me down,” he muttered. After a moment he turned back to his list of questions.